

Problem

Design a problem to help other students better understand mesh analysis.

Step-by-step solution

Step 1 of 6

Consider the following frequency domain circuit for better understand and the procedure to calculate the mesh currents using mesh analysis.

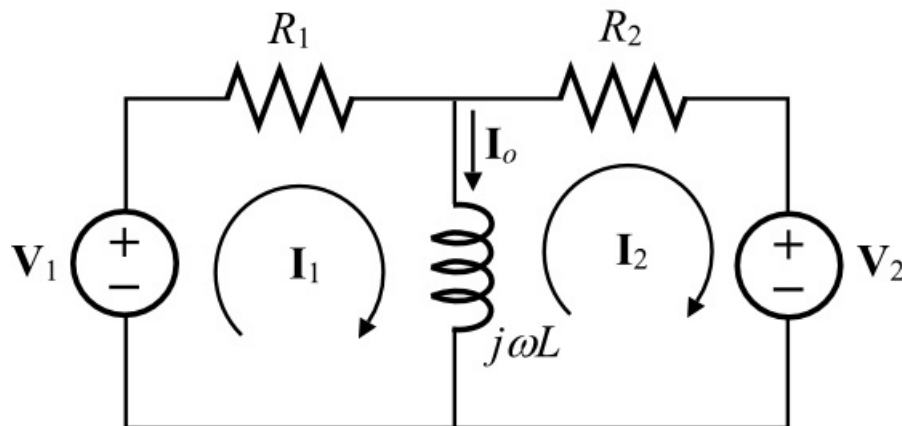


Figure 1

Step 2 of 6

From the circuit, the current $\mathbf{I}_o$ is,

$$\mathbf{I}_o = \mathbf{I}_1 - \mathbf{I}_2 \quad \text{..... (1)}$$

Apply Kirchhoff's voltage law to mesh 1.

$$-\mathbf{V}_1 + \mathbf{I}_1 R_1 + j\omega L (\mathbf{I}_1 - \mathbf{I}_2) = 0$$

$$(R_1 + j\omega L) \mathbf{I}_1 - (j\omega L) \mathbf{I}_2 = \mathbf{V}_1 \quad \text{..... (2)}$$

Apply Kirchhoff's voltage law to mesh 2.

$$j\omega L (\mathbf{I}_2 - \mathbf{I}_1) + \mathbf{I}_2 R_2 + \mathbf{V}_2 = 0$$

$$-(j\omega L) \mathbf{I}_1 + (R_2 + j\omega L) \mathbf{I}_2 = -\mathbf{V}_2 \quad \text{..... (3)}$$

Step 3 of 6

Write equation (2) and equation (3) in matrix form.

$$\begin{bmatrix} R_1 + j\omega L & -j\omega L \\ -j\omega L & R_2 + j\omega L \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{V}_1 \\ -\mathbf{V}_2 \end{bmatrix}$$

Calculate Δ .

$$\begin{aligned} \Delta &= \begin{vmatrix} R_1 + j\omega L & -j\omega L \\ -j\omega L & R_2 + j\omega L \end{vmatrix} \\ &= (R_1 + j\omega L)(R_2 + j\omega L) - (-j\omega L)(-j\omega L) \\ &= R_1 R_2 + j\omega R_2 L + j\omega R_1 L - \omega^2 L^2 + \omega^2 L^2 \\ &= R_1 R_2 + j\omega L(R_1 + R_2) \end{aligned}$$

Step 4 of 6

The matrix for Δ_1 is,

$$\Delta_1 = \begin{bmatrix} \mathbf{V}_1 & -j\omega L \\ -\mathbf{V}_2 & R_2 + j\omega L \end{bmatrix}$$

Calculate determinant of Δ_1 .

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} \mathbf{V}_1 & -j\omega L \\ -\mathbf{V}_2 & R_2 + j\omega L \end{vmatrix} \\ &= (\mathbf{V}_1)(R_2 + j\omega L) - (-\mathbf{V}_2)(-j\omega L) \\ &= (R_2 + j\omega L)\mathbf{V}_1 - (j\omega L)\mathbf{V}_2 \end{aligned}$$

The matrix for Δ_2 is,

$$\Delta_2 = \begin{bmatrix} R_1 + j\omega L & \mathbf{V}_1 \\ -j\omega L & -\mathbf{V}_2 \end{bmatrix}$$

Calculate determinant of Δ_2 .

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} R_1 + j\omega L & \mathbf{V}_1 \\ -j\omega L & -\mathbf{V}_2 \end{vmatrix} \\ &= (R_1 + j\omega L)(-\mathbf{V}_2) - (-j\omega L)(\mathbf{V}_1) \\ &= (j\omega L)\mathbf{V}_1 - (R_1 + j\omega L)\mathbf{V}_2 \end{aligned}$$

Step 5 of 6

Calculate the mesh current $\mathbf{I}_1$.

$$\mathbf{I}_1 = \frac{\Delta_1}{\Delta}$$

Substitute $R_1 R_2 + j\omega L(R_1 + R_2)$ for Δ and $(R_2 + j\omega L)\mathbf{V}_1 - (j\omega L)\mathbf{V}_2$ for Δ_1 .

$$\mathbf{I}_1 = \frac{(R_2 + j\omega L)\mathbf{V}_1 - (j\omega L)\mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)}$$

Calculate the mesh current $\mathbf{I}_2$.

$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta}$$

Substitute $R_1 R_2 + j\omega L(R_1 + R_2)$ for Δ and $(j\omega L)\mathbf{V}_1 - (R_1 + j\omega L)\mathbf{V}_2$ for Δ_2 .

$$\mathbf{I}_2 = \frac{(j\omega L)\mathbf{V}_1 - (R_1 + j\omega L)\mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)}$$

Step 6 of 6

Calculate the current $\mathbf{I}_o$.

$$\mathbf{I}_o = \mathbf{I}_1 - \mathbf{I}_2$$

Substitute $\frac{(R_2 + j\omega L)\mathbf{V}_1 - (j\omega L)\mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)}$ for $\mathbf{I}_1$ and $\frac{(j\omega L)\mathbf{V}_1 - (R_1 + j\omega L)\mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)}$ for $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{I}_o &= \frac{(R_2 + j\omega L)\mathbf{V}_1 - (j\omega L)\mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)} - \frac{(j\omega L)\mathbf{V}_1 - (R_1 + j\omega L)\mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)} \\ &= \frac{(R_2 + j\omega L)\mathbf{V}_1 - (j\omega L)\mathbf{V}_2 - (j\omega L)\mathbf{V}_1 + (R_1 + j\omega L)\mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)} \\ &= \frac{(R_2 + j\omega L - j\omega L)\mathbf{V}_1 - (j\omega L - R_1 - j\omega L)\mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)} \\ &= \frac{R_2 \mathbf{V}_1 + R_1 \mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)}\end{aligned}$$

Therefore, the current $\mathbf{I}_o$ is $\boxed{\frac{R_2 \mathbf{V}_1 + R_1 \mathbf{V}_2}{R_1 R_2 + j\omega L(R_1 + R_2)}}$.